Shubham Mirg

Contact Information

Shortlidge Road,
329 CBEB,
University Park 16802,
PA, USA

smirg@psu.edu
shubham.mirg@gmail.com

Google Scholar • ORCID: 0000-0003-1811-6047

Research Interests

Ultrasound neuromodulation, Novel ultrasound stimulation devices, Neuroimaging, Multimodal imaging platforms, Photoacoustic imaging, Functional ultrasound imaging, Ultrasound localization microscopy

Education

Pennsylvania State University, Pennsylvania, USA

Ph.D., Biomedical Engineering, August 2025

- Advisor: Sri-Rajasekhar Kothapalli, Ph.D
- *GPA: 3.91*

Indian Institute of Technology Kanpur, Kanpur, UP

M.S.(Research), Photonics, January 2020

- Thesis Topic: *A Study of Fiber-Optic Interferometric Hydrophone Sensor Array with Homodyne PGC scheme in Conjunction with TDM Approach*
- Advisor: Pradeep Kumar K., Ph.D
- *CPI: 9.5*

Netaji Subhas Institute of Technology, University of Delhi, New Delhi, Delhi

B.E., Electronics and Communications Engineering, May 2015

- Thesis Topic: *Chaotic signals and their Applications in Digital Communication*
- Advisors: Sujata Sengar, Ph.D and Shree Prakash Singh, Ph.D
- Graduated with *First Class* and a *CGPA of 7.516*

Academic Projects

Simultaneous Functional Ultrasound, Intrinsic Optical Signal, and Widefield Calcium Neuroimaging 2025-2026

- Developed a trimodal imaging platform combining functional ultrasound (fUS), intrinsic optical signal (IOS), and widefield calcium imaging to simultaneously capture cerebral hemodynamics and neural activity in the awake and anaesthetized rodent brain.
- Established awake-mouse functional ultrasound (fUS) imaging as a capability for the lab.
- Built the acquisition system on Verasonics Vantage and Vantage NXT systems, using Arduino-based TTL signals to synchronize ultrasound, imaging and behavioral cameras.
- Developed whisker and visual stimulus delivery for sensory-evoked imaging experiments.
- Wrote MATLAB scripts for data acquisition and analysis.
- Designed a 3D-printed probe holder to integrate the ultrasound and optical imaging pathways.
- Cross-validated the fUS signal against intrinsic hemoglobin signals to characterize its sensitivity and linearity.
- Derived transfer functions for fUS and compared them against hemoglobin-based measures.
- Demonstrated the platform's utility for multiscale imaging in a glioblastoma (GBM) mouse model.
- Supervisor: Sri-Rajasekhar Kothapalli

Transparent Ultrasound Transducer Cranial Window for Ultrasound Neuromodulation and Imaging 2023-2025

- Investigated a transparent ultrasound transducer platform to study the effects of ultrasound neuromodulation using multimodal in vivo imaging.
- Proposed a transparent ultrasound transducer based cranial window over thinned skull for the awake mouse brain, enabling multimodal optical and photoacoustic imaging.
- Developed the surgical preparation and cranial window protocol for chronic awake-mouse imaging.
- Built optical imaging setups for intrinsic optical signal and widefield calcium imaging.
- Assessed hemodynamic parameters across a range of ultrasound stimulation conditions and demonstrated neurovascular-coupling-driven responses.
- Supervisor: Sri-Rajasekhar Kothapalli

A Multi-Parametric Ultrafast Ultrasound and Photoacoustic Imaging Platform for Mice Brain Imaging 2022-2024

- Contributed to a combined multi-parametric ultrafast ultrasound and photoacoustic imaging platform for assessing cerebral hemodynamic and oxygen saturation changes in rodent brains, developing ultrasound localization imaging methods, code, and data analysis.
- Worked on ultrasound localization microscopy (ULM) data analysis applied to a glioblastoma (GBM) mouse model.
- Supervisor: Sri-Rajasekhar Kothapalli

Development of Transparent Ultrasound Transducers 2021-2025

- Proposed the use of lead magnesium niobate-lead titanate (PMN-PT) piezoelectric material to improve transparent ultrasound transducer sensitivity, contributing to transducer fabrication and characterization.
- Characterized transducer performance and noise through pulse-echo, hydrophone pressure, and photoacoustic measurements.
- Compared PMN-PT against lithium niobate-based ultrasound transducers, using PiezoCAD for modeling and MATLAB for data analysis.
- Demonstrated that PMN-PT is a viable alternative to lithium niobate for developing transparent ultrasound transducers in multimodal optical and photoacoustic imaging setups.
- Designed an electrical impedance matching circuit to improve transparent ultrasound transducer performance.
- Supervisor: Sri-Rajasekhar Kothapalli

Fiber-Optic Interferometer Hydrophone Sensor Array 2016-2020

- Investigated the two main sensor array design considerations: passive interrogation schemes and multiplexing approaches.
- Experimentally implemented a homodyne phase-generated carrier for the passive interrogation scheme, demonstrating successful recovery of the acoustic signal.
- Proposed and experimentally demonstrated a passive path imbalance measurement technique for interferometers using the same homodyne PGC setup, with signal processing in MATLAB.
- Multiplexed a five-sensor interferometer array using Time Division Multiplexing.
- Part of Master's Thesis.
- Supervisor: Pradeep Kumar K., Ph.D

Chaos in Erbium-Doped Fiber Ring Lasers 2016-2017

- Experimentally demonstrated wide-bandwidth chaos in symmetric dual-port erbium-doped fiber ring lasers.
- Implemented Rosenstein's algorithm in MATLAB to confirm the presence of chaotic dynamics in the ring laser.
- Supervisor: Pradeep Kumar K., Ph.D

Digital Communication Using Chaotic Signals 2014-2015

- Implemented various modulation and demodulation schemes using chaotic signals in MATLAB.
- Evaluated the schemes based on their jamming performance.
- Part of Bachelor's Thesis Project.
- Supervisors: Sujata Sengar, Ph.D and Shree Prakash Singh, Ph.D

NSIT Solar Car Concept 2012-2015

- Member of the electrical design and development team for a solar electric vehicle.
- Built the battery pack and its management system, and integrated the solar panels and DC motor.
- Supervisor: Ranjana Jha, Ph.D

Research Experience

Cross-Disciplinary Neural Engineering T32 Fellow July 2023 to May 2025

Funding Agency: T32 grant from the National Institute of Neurological Disorders and Stroke, an institute within the U.S. National Institutes of Health (T32NS115667)

Senior Student Research Associate May 2019 to July 2019

Project: Single-carrier decoy-state frequency-coded quantum key distribution over 50 km optical fiber
Funding Agency: Department of Science & Technology, India
Supervisor: Pradeep Kumar K., Ph.D

Senior Student Research Associate August 2016 to June 2018

Project: Fiber-Optic Hydrophone Sensor Array for Ultrasound Detection
Supervisor: Pradeep Kumar K., Ph.D

Teaching Experience

Teaching Assistant Spring 2022, 2023, 2026

BME 403 - Biomedical Instrumentation Laboratory,
Instructor: Xiao Liu, Ph.D
Department of Biomedical Engineering,
Pennsylvania State University

Teaching Assistant Fall 2021, 2022

BME 301 - Analysis of Physiological Systems,
Instructor: Sri-Rajasekhar Kothapalli, Ph.D
Department of Biomedical Engineering,
Pennsylvania State University

Teaching Assistant February 2019 to April 2019

NOC19-EE21 - Electromagnetic Waves in Guided and Wireless Media
in National Programme on Technology Enhanced Learning,
Instructor: Pradeep Kumar K., Ph.D
Department of Electrical Engineering,
Indian Institute of Technology Kanpur

Teaching Assistant July 2018 to October 2018

NOC18-EE28 - Fiber-Optic Communication Systems and Techniques
in National Programme on Technology Enhanced Learning,
Instructor: Pradeep Kumar K., Ph.D
Department of Electrical Engineering,
Indian Institute of Technology Kanpur

- Gave lecture on *Pulse Propagation in Optical fibers* as part of the massive open online course lecture series
- Lab Demonstration of fiber optic devices as part of the massive open online course lecture series

Teaching Assistant January 2018 to May 2018

NOC18-EE04 - Electromagnetic Theory
in National Programme on Technology Enhanced Learning,
Instructor: Pradeep Kumar K., Ph.D
Department of Electrical Engineering,
Indian Institute of Technology Kanpur

Awards

- Best Poster Award, 2024 In Vivo Ultrasound Imaging Gordon Research Conference (GRC) 2024
- Penn State Neuroscience Institute Conference Travel Grant 2024
- Freiburg Rising Star Fellowship Award, University of Freiburg, Germany 2024
- Cross-disciplinary Neural Engineering (CDNE) training fellow, National Institutes of Health (T32NS115667) 2024
- John C. and Joanne H. Villforth Graduate Scholarship in Bioengineering, Pennsylvania State University 2024
- Trainee Professional Development Award, Society for Neuroscience 2023
- Best Paper Award Runner-Up, IEEE Sensors Conference 2022
- Travel grant, Department of EE, Indian Institute of Technology Kanpur 2019
- Travel grant, International Conference on Photonics, Optics and Laser Technology 2019
- Academic Excellence Award, Indian Institute of Technology Kanpur 2016

Refereed Journal Publications

10 peer-reviewed journal articles (4 first-author, 2 co-first) and 10 conference proceedings and abstracts.

Mirg, S., Gaddale, P., Kumar, A., Samanta, K., Saini, B., Patil, S.P., Vargas, A.A., Laliwala, A., Exner, A.A., Wang, Y. and Sipe, G.O. (2026). Simultaneous Functional Ultrasound, Intrinsic Optical Signal and Widefield Calcium Neuroimaging. bioRxiv, pp.2026-06. (In Review)

Mirg, S., Samanta, K., Chen, H., Jiang, J., Turner, K.L., Salehi, F., Ramiah, K.M., Drew, P.J. and Kothapalli, S.R. (2026). Integrated ultrasound stimulation and optical imaging of awake mice brain using a transparent ultrasound transducer cranial window. npj Acoustics, 2(1), p.1.

Chen, H., **Mirg, S.**, Gaddale, P., Agrawal, S., Li, M., Nguyen, V., Xu, T., Li, Q., Liu, J., Tu, W., Liu, X., Drew, P.J., Zhang, N., Gluckman, B.J. and Sri-Rajasekhar Kothapalli (2024). Multiparametric Brain Hemodynamics Imaging Using a Combined Ultrafast Ultrasound and Photoacoustic System. Advanced Science, 11(31). doi: 10.1002/advs.202401467.

Lee, K., Davis, B., Wang, X., **Mirg, S.**, Wen, C., Abune, L., ...Wang, Y. (2023). Nanoparticle-Decorated Biomimetic Extracellular Matrix for Cell Nanoencapsulation and Regulation. Angew. Chem. Int. Ed., e202306583.

Mirg, S., Turner, K. L., Chen, H., Drew, P. J., and Kothapalli, S.-R. (2022). Photoacoustic imaging for microcirculation. Microcirculation, 29(6-7), e12776.

Osman, M. S., Chen, H., Creamer, K., Minotto, J., Liu, J., **Mirg, S.,** ...Kothapalli, S.-R. (2022). A Novel Matching Layer Design for Improving the Performance of Transparent Ultrasound Transducers. IEEE Trans. Ultrason. Ferroelectr. Freq. Control, 69(9), 2672–2680.

Chen, H., Agrawal, S., Osman, M., Minotto, J., **Mirg, S.** Liu, J., ...Kothapalli, S.-R. (2022). A Transparent Ultrasound Array for Real-Time Optical, Ultrasound, and Photoacoustic Imaging. BME Front., 2022.

Mirg, S*., Chen, H*., Chen, H., Turner, K. L., Gheres, K. W., ...Kothapalli, S.-R. (2022). Awake mouse brain photoacoustic and optical imaging through a transparent ultrasound cranial window. Opt. Lett., 47(5), 1121–1124. (*** Co-first authors**)

Chen, H.*, **Mirg, S.***, Osman, M., Agrawal, S., Cai, J., Biskowitz, R., ...Kothapalli, S.-R. (2021). A High Sensitivity Transparent Ultrasound Transducer Based on PMN-PT for Ultrasound and Photoacoustic Imaging. IEEE Sens. Lett., 5(11), 1–4. (*** Co-first authors**)

Mirg, S., Jain, A., Pandey, A., Krishnamurthy, P. K., and Landais, P. (2017). Experimental demonstration of 12.5 GHz wideband chaos in symmetric dual-port EDFRL. Appl. Opt., 56(28), 7939–7943.

Conference Publications

Mirg, S., Gaddale, P., Kumar, A., Samanta, K., Saini, B., Patil, S.P., Vargas, A.A., Laliwala, A., Exner, A.A., Wang, Y. and Sipe, G.O. (2026). Simultaneous Functional Ultrasound, Intrinsic Optical Signal and Widefield Calcium Neuroimaging. In Vivo Ultrasound Imaging Gordon Research Conference, 2026. (Poster)

Samanta, K., **Mirg, S.**, Abbasi, A., Mansour, Y., Ramiah, K.M., Jiang, J., Gomez, E.W. and Kothapalli, S.R. (2026). A novel optical elastography platform integrating transparent ultrasound transducer and digital holography. In Optical Elastography and Tissue Biomechanics XIII (Vol. 13853, pp. 73–79). SPIE.

Mirg, S., Samanta, K., Gaddale, P., Salehi, F., Ramiah, M. K., Hyunseok, L., Drew, P. J., and Kothapalli, S. R. (2025). Integrated Ultrasound Neuromodulation and Optical Neuroimaging in Awake Mice Using a Transparent Ultrasound Transducer Cranial Window. Big Ten Neuroscience Annual Meeting, 2025. (Talk and Poster)

Mirg, S., Nudell, V., Salehi, F., Brockway, D. F., Turner, K. L., Crowley, N. A., Drew, P. J., and Kothapalli, S. R. (2024). Transparent ultrasound transducer platform enables multimodal in-vitro and in-vivo investigations of ultrasound neuromodulation. In Vivo Ultrasound Imaging Gordon Research Conference, 2024. (Poster, **Best Poster Award**)

Mirg, S., Mitra, M., Jiang, J., Gaddale, P., Minteer, J., Du, Y., He, P., and Kothapalli, S. R. (2024). Functional ultrasound and Photoacoustic platform for Imaging Carotid Hemodynamics in Atherosclerosis Mice Model. In Vivo Ultrasound Imaging Gordon Research Seminar, 2024. (Poster)

Minteer, J., Du, Y., **Mirg, S.**, Gandhi, A., Rathnam Paluri, Jiang, Y., Ding, W., Sri-Rajasekhar Kothapalli, Bagchi, P., Kim, Y., and He, P. (2025). Atherosclerosis-induced vascular contributions to cognitive impairment. Physiology, 40(S1). doi: 10.1152/physiol.2025.40.s1.0938 (Poster)

Doody, N*., **Mirg, S*.**, Chen, H., Khandare, S., Mitra, M., and Kothapalli, S.-R. (2024). Improved performance of piezoelectric transparent ultrasound transducers using electrical impedance matching circuit. Proceedings Volume 12842, Photons Plus Ultrasound: Imaging and Sensing 2024. SPIE. doi: 10.1117/12.3003083 (Poster, *** Co-first authors**)

Mirg, S., Chen, H., Khandare, S., Osman, M., and Kothapalli, S.-R. (2022). Noise considerations in piezoelectric transparent ultrasound transducers for photoacoustic imaging applications. Proceedings Volume 11960, Photons Plus Ultrasound: Imaging and Sensing 2022. (Poster)

Chen, H., Osman, M., **Mirg, S.**, Agrawal, S., Cai, J., Dangi, A., and Kothapalli, S.-R. (2021). Transparent ultrasound transducers for multiscale photoacoustic imaging. Proceedings Volume 11642, Photons Plus Ultrasound: Imaging and Sensing 2021. SPIE. doi: 10.1117/12.2579056 (Poster)

Mirg, S. and Krishnamurthy, P.K. (2019). Passive Path Imbalance Measurement in Fiber Optic Interferometer using Homodyne PGC Scheme. In Proceedings of the 7th International Conference on Photonics, Optics and Laser Technology: PHOTOPTICS, ISBN 978-989-758-364-3, pages 73–78. (Talk)

Skills

Pre-clinical & Imaging Skills

- Chronic thinned-skull and craniotomy surgical window preparations in rodents
- Functional ultrasound (fUS) and ultrasound localization microscopy (ULM) imaging in awake and anaesthetized rodents
- Intrinsic optical signal, widefield calcium, and laser speckle contrast imaging in awake and anaesthetized rodents
- Rodent ultrasound brain stimulation
- Tail-vein, retro-orbital, intraperitoneal, and subcutaneous injections; perfusion procedures
- Rodent handling and habituation

Hardware Skills

- Transducer fabrication and characterization (pulse-echo, hydrophone, photoacoustic)
- Building optical, ultrasound and photoacoustic imaging setups
- Interfacing and synchronizing cameras (sCMOS/CMOS), generators, and instruments via Arduino TTL triggering
- Fiber-optic systems and interferometry
- Electrical impedance matching circuit design
- 3D-printed component design and prototyping
- High-speed ADCs and DACs (Verasonics, Gage Card), Arduino, Raspberry Pi
- Electrical circuits and soldering
- Familiar with PCB fabrication and FPGA programming

Software & Analysis Skills

- Ultrasound image analysis: Verasonics programming, beamforming, power-Doppler, fUS and ULM processing
- Optical and calcium imaging analysis; image registration and motion correction
- Signal processing, transfer-function analysis, and statistical data analysis
- Proficient: Matlab, Simulink, PiezoCAD, Adobe Illustrator
- Familiar with: Python, Fusion 360, Git, VHDL, Altium Designer, Eagle, SPICE